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Hydrogen peroxide

Hydrogen peroxide is a chemical compound with the formula ‎H₂O₂‎. In its pure form, it is a very pale blue^[5] liquid that is slightly more viscous than water. It is used as an oxidizer, bleaching agent, and antiseptic, usually as a dilute solution (3%–6% by weight) in water for consumer use, and in higher concentrations for industrial use. Concentrated hydrogen peroxide, or "high-test peroxide", decomposes explosively when heated and has been used both as a monopropellant and an oxidizer in rocketry.^[6]

Hydrogen peroxide is a reactive oxygen species and the simplest peroxide, a compound having an oxygen–oxygen single bond. It decomposes slowly into water and elemental oxygen when exposed to light, and rapidly in the presence of organic or reactive compounds. It is typically stored with a stabilizer in a weakly acidic solution in a dark bottle to block light. Hydrogen peroxide is found in biological systems including the human body. Enzymes that use or decompose hydrogen peroxide are classified as peroxidases.

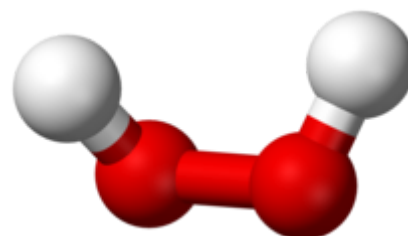
Properties

The boiling point of ‎H₂O₂‎ has been extrapolated as being 150.2 °C (302.4 °F), approximately 50 °C (90 °F) higher than water. In practice, hydrogen peroxide will undergo potentially explosive thermal decomposition if heated to this temperature. It may be safely distilled at lower temperatures under reduced pressure.^[7]

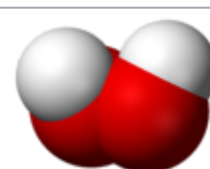
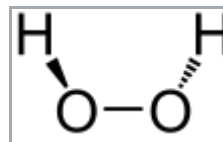
Structure

Hydrogen peroxide (‎H₂O₂‎) is a nonplanar molecule with (twisted) C₂ symmetry; this was first shown by Paul-Antoine Giguère in 1950 using infrared spectroscopy.^{[8][9]} Although the O–O bond is a single bond, the molecule has a relatively high rotational barrier of 386 cm^{−1} (4.62 kJ/mol) for rotation between enantiomers via the trans configuration, and 2460 cm^{−1} (29.4 kJ/mol) via the cis configuration.^[10] These barriers are proposed to be due to repulsion between the lone

Hydrogen peroxide



Ball stick model of the hydrogen peroxide molecule



Names

IUPAC name

Hydrogen peroxide

Other names

Dioxidane
Oxidanyl
Perhydroxic acid
O-hydroxyol
Dihydrogen dioxide
Oxygenated water
Peroxaan

Identifiers

CAS Number

7722-84-1 (https://commonchemistry.cas.org/detail?cas_rn=7722-84-1) ✓

3D model (JSmol)

Interactive image
(<https://chemapps.stolaf.edu/jmol/jmol.php?model=OO>)

ChEBI

CHEBI:16240 (<https://www.ebi.ac.uk/chebi/searchId.d>)

pairs of the adjacent oxygen atoms and dipolar effects between the two O–H bonds. For comparison, the rotational barrier for ethane is 1040 cm^{-1} (12.4 kJ/mol).

The approximately 100° dihedral angle between the two O–H bonds makes the molecule chiral. It is the smallest and simplest molecule to exhibit enantiomerism. It has been proposed that the enantiospecific interactions of one rather than the other may have led to amplification of one enantiomeric form of ribonucleic acids and therefore an origin of homochirality in an RNA world.^[11]

The molecular structures of gaseous and crystalline H_2O_2 are significantly different. This difference is attributed to the effects of hydrogen bonding, which is absent in the gaseous state.^[12] Crystals of H_2O_2 are tetragonal with the space group D_4^4 or $P4_12_12$.^[13]

Aqueous solutions

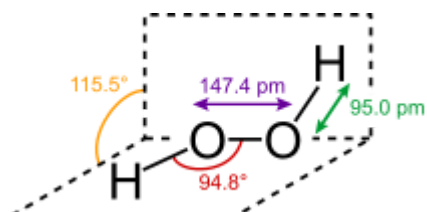
In aqueous solutions, hydrogen peroxide differs from the pure substance due to the effects of hydrogen bonding between water and hydrogen peroxide molecules. Hydrogen peroxide and water form a eutectic mixture, exhibiting freezing-point depression down as low as -56°C ; pure water has a freezing point of 0°C and pure hydrogen peroxide of -0.43°C . The boiling point of the same mixtures is also depressed in relation with the mean of both boiling points (125.1°C). It occurs at 114°C . This boiling point is 14°C greater than that of pure water and 36.2°C less than that of pure hydrogen peroxide.^[14]

	o?chebld=16240 ✓
<u>ChEMBL</u>	ChEMBL71595 (https://www.ebi.ac.uk/chembl/index.php/compound/inspect/ChEMBL71595) ✓
<u>ChemSpider</u>	763 (https://www.chemspider.com/Chemical-Structure.763.html) ✓
<u>ECHA InfoCard</u>	100.028.878 (https://echa.europa.eu/substance-information/-/substanceinfo/100.028.878)
<u>EC Number</u>	231-765-0
<u>IUPHAR/BPS</u>	2448 (http://www.guidetopharmacology.org/GRAC/LigandDisplayForward?tab=summary&ligandId=2448)
<u>KEGG</u>	D00008 (https://www.kegg.jp/entry/D00008) ✓
<u>PubChem CID</u>	784 (https://pubchem.ncbi.nlm.nih.gov/compound/784)
<u>RTECS number</u>	MX0900000 (>90% soln.) MX0887000 (>30% soln.)
<u>UNII</u>	BBX060AN9V (https://precision.fda.gov/uniisearch/srs/unii/BBX060AN9V) ✓
<u>UN number</u>	2015 (>60% soln.) 2014 (20–60% soln.)

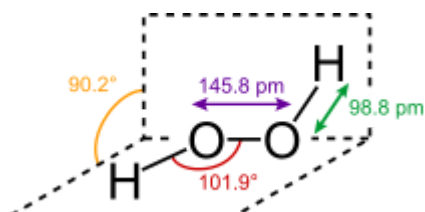
	2984 (8–20% soln.)
CompTox Dashboard (EPA)	DTXSID2020715 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID2020715)
<u>InChI</u> InChI=1S/H2O2/c1-2/h1-2H ✓ Key: MHAJPDPJQMAIIY-UHFFFAOYS A-N ✓ InChI=1/H2O2/c1-2/h1-2H Key: MHAJPDPJQMAIIY-UHFFFAOYA L	
<u>SMILES</u> OO	
Properties	
<u>Chemical formula</u>	H ₂ O ₂
<u>Molar mass</u>	34.0147 g/mol
<u>Appearance</u>	Very light blue liquid
<u>Odor</u>	slightly sharp
<u>Density</u>	1.11 g/cm ³ (20 °C, 30% (w/w) solution) ^[1] 1.450 g/cm ³ (20 °C, pure)
<u>Melting point</u>	−0.43 °C (31.23 °F; 272.72 K)
<u>Boiling point</u>	150.2 °C (302.4 °F; 423.3 K) (decomposes)
<u>Solubility in water</u>	<u>Miscible</u>
<u>Solubility</u>	soluble in <u>ether</u> , <u>alcohol</u> insoluble in petroleum ether
<u>log <i>P</i></u>	−0.43 ^[2]
<u>Vapor pressure</u>	5 mmHg (30 °C) ^[3]
<u>Acidity (p<i>K</i>_a)</u>	11.75

<u>Magnetic susceptibility (χ)</u>	$-17.7 \cdot 10^{-6}$ cm ³ /mol
<u>Refractive index (n_D)</u>	1.4061
<u>Viscosity</u>	1.245 cP (20 °C)
<u>Dipole moment</u>	2.26 D
Thermochemistry	
<u>Heat capacity (C)</u>	1.267 J/(g·K) (gas) 2.619 J/(g·K) (liquid)
<u>Std enthalpy of formation ($\Delta_f H^\ominus_{298}$)</u>	-187.80 kJ/mol
Pharmacology	
<u>ATC code</u>	A01AB02 (WHO (https://www.whocc.no/atc_ddd_index/?code=A01AB02)) D08AX01 (WHO (https://www.whocc.no/atc_ddd_index/?code=D08AX01))), D11AX25 (WHO (https://www.who.c.no/atc_ddd_index/?code=D11AX25))), S02AA06 (WHO (https://www.whocc.no/atc_ddd_index/?code=S02AA06)))
Hazards	
GHS labelling:	
<u>Pictograms</u>	
<u>Signal word</u>	Danger
<u>Hazard statements</u>	H271, H302, H314, H332, H335, H412

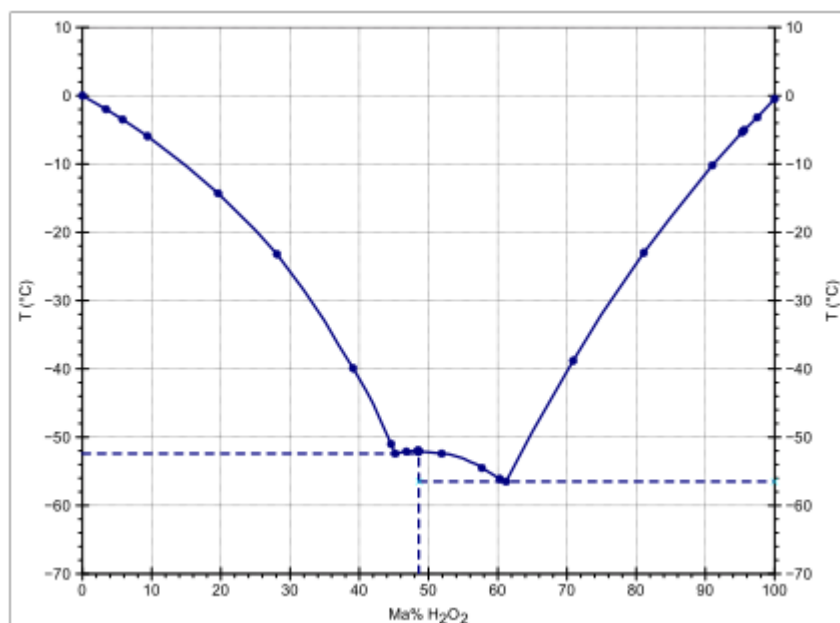
<u>Precautionary statements</u>	P280, P305+P351+P338, P310
NFPA 704 (fire diamond)	
<u>Flash point</u>	Non-flammable
Lethal dose or concentration (LD, LC):	
LD ₅₀ (<u>median dose</u>)	1518 mg/kg 2000 mg/kg (oral, mouse) ^[4]
LC ₅₀ (<u>median concentration</u>)	1418 ppm (rat, 4 hr) ^[4]
LC _{Lo} (<u>lowest published</u>)	227 ppm (mouse) ^[4]
NIOSH (US health exposure limits):	
<u>PEL</u> (Permissible)	TWA 1 ppm (1.4 mg/m ³) ^[3]
<u>REL</u> (Recommended)	TWA 1 ppm (1.4 mg/m ³) ^[3]
<u>IDLH</u> (Immediate danger)	75 ppm ^[3]
<u>Safety data sheet</u> (SDS)	ICSC 0164 (>60% soln.) (https://www.inchem.org/documents/icsc/icsc/eics0164.htm)
Related compounds	
Related compounds	<u>Water</u> <u>Ozone</u> <u>Hydrazine</u> <u>Hydrogen disulfide</u> <u>Dioxygen difluoride</u>
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at 25 °C [77 °F], 100 kPa). ✓ <u>verify</u> (what is ✓✗ ?) <u>Infobox references</u>	



Structure and dimensions of H_2O_2 in the gas phase



Structure and dimensions of H_2O_2 in the solid (crystalline) phase



Phase diagram of H_2O_2 and water: Area above blue line is liquid. Dotted lines separate solid-liquid phases from solid-solid phases.

Density of aqueous solution of H_2O_2

H_2O_2 (w/w)	Density (g/cm^3)	Temp. ($^\circ\text{C}$)
3%	1.0095	15
27%	1.10	20
35%	1.13	20
50%	1.20	20
70%	1.29	20
75%	1.33	20
96%	1.42	20
98%	1.43	20
100%	1.45	20

Comparison with analogues

Hydrogen peroxide has several structural analogues with $\text{H}_m\text{X}-\text{XH}_n$ bonding arrangements (water also shown for comparison). It has the highest (theoretical) boiling point of this series ($\text{X} = \text{O}, \text{S}, \text{N}, \text{P}$). Its melting point is also fairly high, being comparable to that of hydrazine and water, with only hydroxylamine crystallising significantly more readily, indicative of particularly strong hydrogen bonding. Diphosphane and hydrogen disulfide exhibit only weak hydrogen bonding and have little chemical similarity to hydrogen peroxide. Structurally, the analogues all adopt similar skewed structures, due to repulsion between adjacent lone pairs.

Properties of H₂O₂ and its analogues
Values marked * are extrapolated

Name	Formula	Molar mass (g/mol)	Melting point (°C)	Boiling point (°C)
<u>Water</u>	HOH	18.02	0.00	99.98
Hydrogen peroxide	HOOH	34.01	−0.43	150.2*
<u>Hydrogen disulfide</u>	HSSH	66.15	−89.6	70.7
<u>Hydrazine</u>	H ₂ NNH ₂	32.05	2	114
<u>Hydroxylamine</u>	NH ₂ OH	33.03	33	58*
<u>Diphosphane</u>	H ₂ PPH ₂	65.98	−99	63.5*

Discovery

Alexander von Humboldt is sometimes said to have been the first to report the first synthetic peroxide, barium peroxide, in 1799 as a by-product of his attempts to decompose air, although this is disputed due to von Humboldt's ambiguous wording.^[15] Nineteen years later Louis Jacques Thénard recognized that this compound could be used for the preparation of a previously unknown compound, which he described as *eau oxygénée* ("oxygenated water") – subsequently known as hydrogen peroxide.^{[16][17][18]} Today, the term "oxygenated water" may appear on retail packaging referring to mixtures containing either water and hydrogen peroxide or water and dissolved oxygen. This could cause personal injury if the difference is not properly understood by the user.^[19]

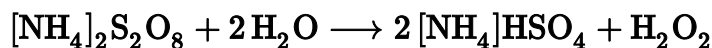
An improved version of Thénard's process used hydrochloric acid, followed by addition of sulfuric acid to precipitate the barium sulfate byproduct. This process was used from the end of the 19th century until the middle of the 20th century.^[20]

The bleaching effect of peroxides and their salts on natural dyes had been known since Thénard's experiments in the 1820s, but early attempts of industrial production of peroxides failed. The first plant producing hydrogen peroxide was built in 1873 in Berlin. The discovery of the synthesis of hydrogen peroxide by electrolysis with sulfuric acid introduced the more efficient electrochemical method. It was first commercialized in 1908 in Weissenstein, Carinthia, Austria. The anthraquinone process, which is still used, was developed during the 1930s by the German chemical manufacturer IG Farben in Ludwigshafen. The increased demand and improvements in the synthesis methods resulted in the rise of the annual production of hydrogen peroxide from 35,000 tonnes in 1950, to over 100,000 tonnes in 1960, to 300,000 tonnes by 1970; by 1998 it reached 2.7 million tonnes.^[21]

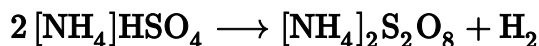
Early attempts failed to produce neat hydrogen peroxide. Anhydrous hydrogen peroxide was first obtained by vacuum distillation.^[22]

Determination of the molecular structure of hydrogen peroxide proved to be very difficult. In 1892, the Italian physical chemist Giacomo Carrara (1864–1925) determined its molecular mass by freezing-point depression, which confirmed that its molecular formula is H₂O₂.^[23] H₂O=O seemed to be just as possible as the modern structure, and as late as in the middle of the 20th century at least half a dozen hypothetical isomeric variants of two main options seemed to be consistent with the available evidence.^[24] In 1934, the English mathematical physicist William Penney and the Scottish physicist Gordon Sutherland proposed a molecular structure for hydrogen peroxide that was very similar to the presently accepted one.^{[25][26]}

Previously, hydrogen peroxide was prepared industrially by hydrolysis of ammonium persulfate:

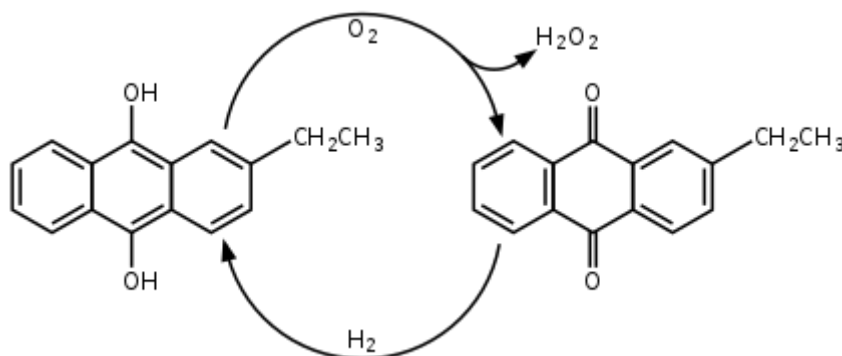


which was itself obtained by the electrolysis of a solution of ammonium bisulfate ($[\text{NH}_4]\text{HSO}_4$) in sulfuric acid:^[27]



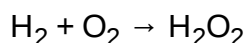
Production

Today, hydrogen peroxide is manufactured almost exclusively by the anthraquinone process, which was originally developed by BASF in 1939. It begins with the reduction of an anthraquinone (such as 2-ethylanthraquinone or the 2-amyl derivative) to the corresponding anthrahydroquinone, typically by hydrogenation on a palladium catalyst. In the presence of oxygen, the anthrahydroquinone then undergoes autoxidation: the labile hydrogen atoms of the hydroxy groups transfer to the oxygen molecule, to give hydrogen peroxide and regenerating the anthraquinone. Most commercial processes achieve oxidation by bubbling compressed air through a solution of the anthrahydroquinone, with the hydrogen peroxide then extracted from the solution and the anthraquinone recycled back for successive cycles of hydrogenation and oxidation.^{[28][29]}



Catalytic cycle for the anthraquinone process to produce hydrogen peroxide: an anthraquinone (right) is reduced using hydrogen to produce the corresponding anthrahydroquinone (left). This is oxidized using oxygen to produce hydrogen peroxide and recover anthraquinone.

The net reaction for the anthraquinone-catalyzed process is :^[28]



The economics of the process depend heavily on effective recycling of the extraction solvents, the hydrogenation catalyst and the expensive quinone.

Other sources

Small, but detectable, amounts of hydrogen peroxide can be formed by several methods. Small amounts are formed by electrolysis of dilute acid around the cathode where hydrogen evolves if oxygen is bubbled around it. It is also produced by exposing water to ultraviolet rays from a mercury lamp, or an electric arc while confining it in a UV transparent vessel (e.g. quartz). It is detectable in ice water after burning a hydrogen gas stream aimed towards it and is also detectable on floating ice. Rapidly cooling humid air blown through an approximately 2,000 °C spark gap results in detectable amounts.^[30]



ISO tank container for hydrogen peroxide transportation

A commercially viable process to produce hydrogen peroxide directly from the environment has been of interest for many years. Efficient direct synthesis is difficult to achieve, as the reaction of hydrogen with oxygen thermodynamically favours production of water. Systems for direct synthesis have been developed, most of which employ finely dispersed metal catalysts similar to those used for hydrogenation of organic substrates.^{[31][32]} One economic obstacle has been that direct processes give a dilute solution uneconomic for transportation. None of these has yet reached a point where it can be used for industrial-scale synthesis.



A tank car designed for transporting hydrogen peroxide by rail

Availability

Hydrogen peroxide is most commonly available as a solution in water. For consumers, it is usually available from pharmacies at 3 and 6 wt% concentrations. The concentrations are sometimes described in terms of the volume of oxygen gas generated; one milliliter of a 20-volume solution generates twenty milliliters of oxygen gas when completely decomposed. For laboratory use, 30 wt% solutions are most common. Commercial grades from 70% to 98% are also available, but due to the potential of solutions of more than 68% hydrogen peroxide to be converted entirely to steam and oxygen (with the temperature of the steam increasing as the concentration increases above 68%) these grades are potentially far more hazardous and require special care in dedicated storage areas. Buyers must typically allow inspection by commercial manufacturers.

In 1994, world production of H_2O_2 was around 1.9 million tonnes and grew to 2.2 million in 2006,^[33] most of which was at a concentration of 70% or less. In that year, bulk 30% H_2O_2 sold for around 0.54 USD/kg, equivalent to US\$1.50/kg (US\$0.68/lb) on a "100% basis".^[28]

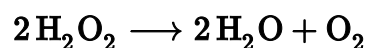
Natural occurrence

Hydrogen peroxide occurs in surface water, in groundwater, and in the atmosphere. It forms upon illumination or natural catalytic action by substances contained in water. Sea water contains 0.5 to 14 $\mu\text{g/L}$ of hydrogen peroxide, and freshwater contains 1 to 30 $\mu\text{g/L}$.^[21] Concentrations in air are about 0.4 to 4 $\mu\text{g/m}^3$, varying over several orders of magnitude depending in conditions such as season, altitude, daylight and water vapor content. In rural nighttime air it is less than 0.014 $\mu\text{g/m}^3$, and in moderate photochemical smog it is 14 to 42 $\mu\text{g/m}^3$.^[34]

Reactions

Decomposition

Hydrogen peroxide decomposes to form water and oxygen with a ΔH° of -2884.5 kJ/kg ^[35] and a ΔS of $70.5 \text{ J/(mol}\cdot\text{K)}$:



The rate of decomposition increases with rise in temperature, concentration, and pH (H_2O_2 being unstable under alkaline conditions), with cool, dilute, and acidic solutions showing the best stability. Decomposition is catalysed by various redox-active ions or compounds, including most transition metals and their

compounds (e.g. manganese dioxide (MnO_2), silver, and platinum).^[36] Certain metal ions, such as Fe^{2+} or Ti^{3+} , can cause the decomposition to take a different path, with free radicals such as the hydroxyl radical ($\text{HO}^\bullet$) and hydroperoxyl ($\text{HOO}^\bullet$) being formed. Potassium iodide (KI) reacts particularly rapidly and forms the basis of the elephant toothpaste demonstration. Hydrogen peroxide can also be decomposed biologically by the enzyme catalase. The decomposition of hydrogen peroxide liberates oxygen and heat; this can be dangerous, as spilling high-concentration hydrogen peroxide on a flammable substance can cause an immediate fire.

Redox reactions

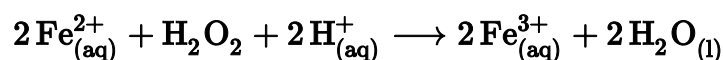
The redox properties of hydrogen peroxide depend on pH as acidic conditions exacerbate the power of oxidizing agents and basic conditions exacerbate the power of reducing agents. As hydrogen peroxide exhibits ambivalent redox properties, being simultaneously an oxidizer or a reductant, its redox behavior immediately depends on pH.

In acidic solutions, H_2O_2 is a powerful oxidizer, stronger than chlorine, chlorine dioxide, and potassium permanganate. When used for cleaning laboratory glassware, a solution of hydrogen peroxide and sulfuric acid is referred to as Piranha solution.

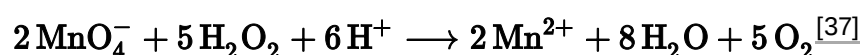
H_2O_2 is a source of hydroxyl radicals ($^\bullet\text{OH}$), which are highly reactive.

<u>Oxidizing reagent</u>	<u>Reduced product</u>	<u>Oxidation potential (V)</u>
F_2	HF	3.0
O_3	O_2	2.1
H_2O_2	H_2O	1.8
KMnO_4	MnO_2	1.7
ClO_2	HClO	1.5
Cl_2	Cl^-	1.4

In acidic solutions, Fe^{2+} is oxidized to Fe^{3+} (hydrogen peroxide acting as an oxidizing agent):

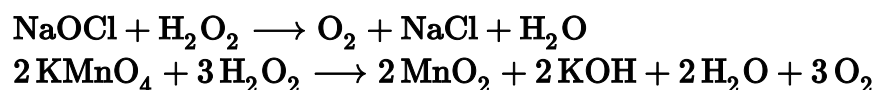


and sulfite (SO_3^{2-}) is oxidized to sulfate (SO_4^{2-}). However, potassium permanganate is reduced to Mn^{2+} by acidic H_2O_2 .



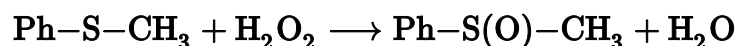
Under alkaline conditions, however, some of these reactions reverse; for example, Mn^{2+} is oxidized to Mn^{4+} (as MnO_2).

In basic solutions, hydrogen peroxide is a strong reductant and can reduce a variety of inorganic ions. When H_2O_2 acts as a reducing agent, oxygen gas is also produced. For example, hydrogen peroxide will reduce sodium hypochlorite and potassium permanganate, which is a convenient method for preparing oxygen in the laboratory:



Organic reactions

Hydrogen peroxide is frequently used as an oxidizing agent. Illustrative is oxidation of thioethers to sulfoxides.^{[38][39]}



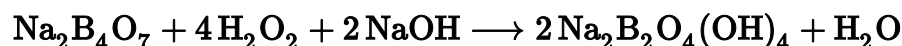
Alkaline hydrogen peroxide is used for epoxidation of electron-deficient alkenes such as acrylic acid derivatives,^[40] and for the oxidation of alkylboranes to alcohols, the second step of hydroboration-oxidation. It is also the principal reagent in the Dakin oxidation process.

Precursor to other peroxide compounds

Hydrogen peroxide is a weak acid, forming hydroperoxide or peroxide salts with many metals.

It also converts metal oxides into the corresponding peroxides. For example, upon treatment with hydrogen peroxide, chromic acid (CrO_3 and H_2SO_4) forms a blue peroxide $\text{CrO}(\text{O}_2)_2$.

This kind of reaction is used industrially to produce peroxoanions. For example, reaction with borax leads to sodium perborate, a bleach used in laundry detergents:

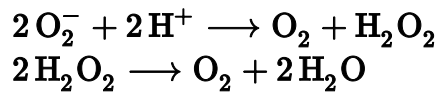


H_2O_2 converts carboxylic acids (RCO_2H) into peroxy acids ($\text{RC}(\text{O})\text{O}_2\text{H}$), which are themselves used as oxidizing agents. Hydrogen peroxide reacts with acetone to form acetone peroxide and with ozone to form trioxidane. Hydrogen peroxide forms stable adducts with urea (Hydrogen peroxide - urea), sodium carbonate (sodium percarbonate) and other compounds.^[41] An acid-base adduct with triphenylphosphine oxide is a useful "carrier" for H_2O_2 in some reactions.

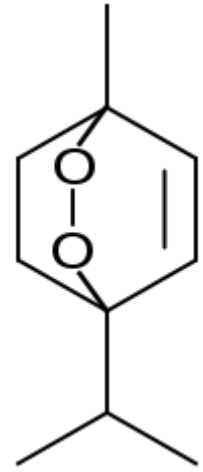
Hydrogen peroxide is both an oxidizing agent and reducing agent. The oxidation of hydrogen peroxide by sodium hypochlorite yields singlet oxygen. The net reaction of a ferric ion with hydrogen peroxide is a ferrous ion and oxygen. This proceeds via single electron oxidation and hydroxyl radicals. This is used in some organic chemistry oxidations, e.g. in the Fenton's reagent. Only catalytic quantities of iron ion is needed since peroxide also oxidizes ferrous to ferric ion. The net reaction of hydrogen peroxide and permanganate or manganese dioxide is manganous ion; however, until the peroxide is spent some manganese ions are reoxidized to make the reaction catalytic. This forms the basis for common monopropellant rockets.

Biological function

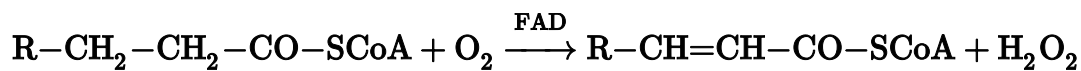
Hydrogen peroxide is formed in humans and other animals as a short-lived product in biochemical processes and is toxic to cells. The toxicity is due to oxidation of proteins, membrane lipids and DNA by the peroxide ions.^[42] The class of biological enzymes called superoxide dismutase (SOD) is developed in nearly all living cells as an important antioxidant agent. They promote the disproportionation of superoxide into oxygen and hydrogen peroxide, which is then rapidly decomposed by the enzyme catalase to oxygen and water.^[43]



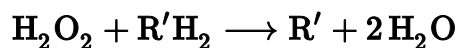
Peroxisomes are organelles found in virtually all eukaryotic cells.^[44] They are involved in the catabolism of very long chain fatty acids, branched chain fatty acids, D-amino acids, polyamines, and biosynthesis of plasmalogens, ether phospholipids critical for the normal function of mammalian brains and lungs.^[45] Upon oxidation, they produce hydrogen peroxide in the following process catalyzed by flavin adenine dinucleotide (FAD):^[46]



Ascaridole

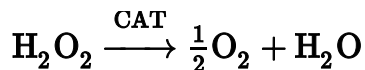


Catalase, another peroxisomal enzyme, uses this H_2O_2 to oxidize other substrates, including phenols, formic acid, formaldehyde, and alcohol, by means of a peroxidation reaction:

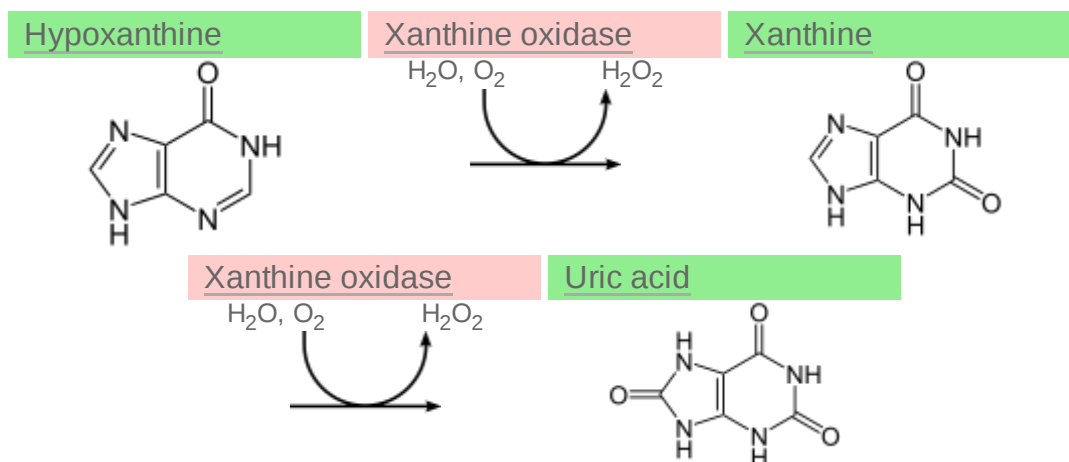


thus eliminating the poisonous hydrogen peroxide in the process.

This reaction is important in liver and kidney cells, where the peroxisomes neutralize various toxic substances that enter the blood. Some of the ethanol humans drink is oxidized to acetaldehyde in this way.^[47] In addition, when excess H_2O_2 accumulates in the cell, catalase converts it to H_2O through this reaction:



Another origin of hydrogen peroxide is the degradation of adenosine monophosphate which yields hypoxanthine. Hypoxanthine is then oxidatively catabolized first to xanthine and then to uric acid, and the reaction is catalyzed by the enzyme xanthine oxidase:^[48]



Degradation of hypoxanthine through xanthine to uric acid to form hydrogen peroxide.

The degradation of guanosine monophosphate yields xanthine as an intermediate product which is then converted in the same way to uric acid with the formation of hydrogen peroxide.^[48]

Eggs of sea urchin, shortly after fertilization by a sperm, produce hydrogen peroxide. It is then quickly dissociated to $\text{HO}\cdot$ radicals. The radicals serve as initiator of radical polymerization, which surrounds the eggs with a protective layer of polymer.^[49]

The bombardier beetle has a device which allows it to shoot corrosive and foul-smelling bubbles at its enemies. The beetle produces and stores hydroquinone and hydrogen peroxide, in two separate reservoirs in the rear tip of its abdomen. When threatened, the beetle contracts muscles that force the two reactants through valved tubes into a mixing chamber containing water and a mixture of catalytic enzymes. When combined, the reactants undergo a violent exothermic chemical reaction, raising the temperature to near the boiling point of water. The boiling, foul-smelling liquid partially becomes a gas (flash evaporation) and is expelled through an outlet valve with a loud popping sound.^{[50][51][52]}



Australian bombardier beetle

Hydrogen peroxide is a signaling molecule of plant defense against pathogens.^[53]

Hydrogen peroxide has roles as a signalling molecule in the regulation of a wide variety of biological processes.^[54] The compound is a major factor implicated in the free-radical theory of aging, based on how readily hydrogen peroxide can decompose into a hydroxyl radical and how superoxide radical byproducts of cellular metabolism can react with ambient water to form hydrogen peroxide.^[55] These hydroxyl radicals in turn readily react with and damage vital cellular components, especially those of the mitochondria.^{[56][57][58]} At least one study has also tried to link hydrogen peroxide production to cancer.^[59] These studies have frequently been quoted in fraudulent treatment claims.

The amount of hydrogen peroxide in biological systems can be assayed using a fluorometric assay.^[60]

Uses

Bleaching

About 60% of the world's production of hydrogen peroxide is used for pulp- and paper-bleaching.^[33] The second major industrial application is the manufacture of sodium percarbonate and sodium perborate, which are used as mild bleaches in laundry detergents. Sodium percarbonate, which is an adduct of sodium carbonate and hydrogen peroxide, is the active ingredient in such laundry products as OxiClean and Tide laundry detergent. When dissolved in water, it releases hydrogen peroxide and sodium carbonate.^[20] By themselves these bleaching agents are only effective at wash temperatures of 60 °C (140 °F) or above and so, often are used in conjunction with bleach activators, which facilitate cleaning at lower temperatures. It has also been used as a flour bleaching agent and a tooth and bone whitening agent.

Production of organic compounds

It is used in the production of various organic peroxides with dibenzoyl peroxide being a high volume example. Peroxy acids, such as peracetic acid and meta-chloroperoxybenzoic acid also are produced using hydrogen peroxide. Hydrogen peroxide has been used for creating organic peroxide-based explosives, such as acetone peroxide. It is used as an initiator in polymerizations.

Sewage treatment

Hydrogen peroxide is used in certain waste-water treatment processes to remove organic impurities. In advanced oxidation processing, the Fenton reaction^{[61][62]} gives the highly reactive hydroxyl radical ($\bullet\text{OH}$). This degrades organic compounds, including those that are ordinarily robust, such as aromatic or halogenated compounds.^[63] It can also oxidize sulfur-based compounds present in the waste; which is beneficial as it generally reduces their odour.^[64]

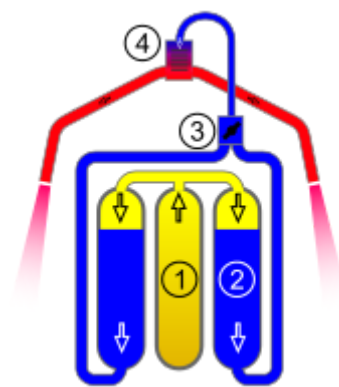
Disinfectant

Hydrogen peroxide may be used for the sterilization of various surfaces,^[65] including surgical tools,^[66] and may be deployed as a vapour (VHP) for room sterilization.^[67] H_2O_2 demonstrates broad-spectrum efficacy against viruses, bacteria, yeasts, and bacterial spores.^{[68][69]} In general, greater activity is seen against Gram-positive than Gram-negative bacteria; however, the presence of catalase or other peroxidases in these organisms may increase tolerance in the presence of lower concentrations.^[70] Lower levels of concentration (3%) will work against most spores; higher concentrations (7 to 30%) and longer contact times will improve sporicidal activity.^{[69][71]}

Hydrogen peroxide is seen as an environmentally safe alternative to chlorine-based bleaches, as it degrades to form oxygen and water and it is generally recognized as safe as an antimicrobial agent by the U.S. Food and Drug Administration (FDA).^[72]

Propellant

High-concentration H_2O_2 is referred to as "high-test peroxide" (HTP). It can be used either as a monopropellant (not mixed with fuel) or as the oxidizer component of a bipropellant rocket. Use as a monopropellant takes advantage of the decomposition of 70–98% concentration hydrogen peroxide into steam and oxygen. The propellant is pumped into a reaction chamber, where a catalyst, usually a silver or platinum screen, triggers decomposition, producing steam at over 600 °C (1,100 °F), which is expelled through a nozzle, generating thrust. H_2O_2 monopropellant produces a maximal specific impulse (I_{sp}) of 161 s (1.6 kN·s/kg). Peroxide was the first major monopropellant adopted for use in rocket applications. Hydrazine eventually replaced hydrogen-peroxide monopropellant thruster applications primarily because of a 25% increase in the vacuum specific impulse.^[73] Hydrazine (toxic) and hydrogen peroxide (less-toxic [ACGIH TLV 0.01 and 1 ppm respectively]) are the only two monopropellants (other than cold gases) to have been widely adopted and utilized for propulsion and power applications. The Bell Rocket Belt, reaction control systems for X-1, X-15, Centaur, Mercury, Little Joe, as well as the turbo-pump gas generators for X-1, X-15, Jupiter, Redstone and Viking used hydrogen peroxide as a monopropellant.^[74] The RD-107 engines (used from 1957 to present) in the R-7 series of rockets decompose hydrogen peroxide to power the turbopumps.



Rocket-belt hydrogen-peroxide propulsion system used in a jet pack

As a bipropellant, H_2O_2 is decomposed to burn a fuel as an oxidizer. Specific impulses as high as 350 s (3.5 kN·s/kg) can be achieved, depending on the fuel. Peroxide used as an oxidizer gives a somewhat lower I_{sp} than liquid oxygen, but is dense, storable, non-cryogenic and can be more easily used to drive gas turbines to give high pressures using an efficient *closed cycle*. It may also be used for regenerative cooling of rocket engines. Peroxide was used very successfully as an oxidizer in World War II German rocket motors (e.g. T-Stoff, containing oxyquinoline stabilizer, for both the Walter HWK 109-500 Starthilfe RATO externally podded monopropellant booster system, and for the Walter HWK 109-509 rocket motor series used for the Me 163B), most often used with C-Stoff in a self-igniting hypergolic combination, and for the low-cost British Black Knight and Black Arrow launchers. Presently, HTP is used on ILR-33 AMBER^[75] and Nucleus^[76] suborbital rockets.

In the 1940s and 1950s, the Hellmuth Walter KG—conceived turbine used hydrogen peroxide for use in submarines while submerged; it was found to be too noisy and require too much maintenance compared to diesel-electric power systems. Some torpedoes used hydrogen peroxide as oxidizer or propellant. Operator error in the use of hydrogen-peroxide torpedoes was named as possible causes for the sinking of HMS Sidon and the Russian submarine Kursk.^[77] SAAB Underwater Systems is manufacturing the Torpedo 2000. This torpedo, used by the Swedish Navy, is powered by a piston engine propelled by HTP as an oxidizer and kerosene as a fuel in a bipropellant system.^{[78][79]}

Household use

Hydrogen peroxide has various domestic uses, primarily as a cleaning and disinfecting agent.

Hair bleaching

Diluted H_2O_2 (between 1.9% and 12%) mixed with aqueous ammonia has been used to bleach human hair. The chemical's bleaching property lends its name to the phrase "peroxide blonde".^[80] Hydrogen peroxide is also used for tooth whitening. It may be found in most whitening toothpastes. Hydrogen peroxide has shown positive results involving teeth lightness and chroma shade parameters.^[81] It works by oxidizing colored pigments onto the enamel where the shade of the tooth may become lighter. Hydrogen peroxide may be mixed with baking soda and salt to make a homemade toothpaste.^[82]

Removal of blood stains

Hydrogen peroxide reacts with blood as a bleaching agent, and so if a blood stain is fresh, or not too old, liberal application of hydrogen peroxide, if necessary in more than single application, will bleach the stain fully out. After about two minutes of the application, the blood should be firmly blotted out.^{[83][84]}

Acne treatment

Hydrogen peroxide may be used to treat acne,^[85] although benzoyl peroxide is a more common treatment.

Oral cleaning agent



Contact lenses soaking in a 3% hydrogen peroxide-based solution. The case includes a catalytic disc which neutralises the hydrogen peroxide over time.

The use of dilute hydrogen peroxide as a oral cleansing agent has been reviewed academically to determine its usefulness in treating gingivitis and plaque. Although there is a positive effect when compared with a placebo, it was concluded that chlorhexidine is a much more effective treatment.^[86]

Niche uses

Glow sticks

Hydrogen peroxide reacts with certain di-esters, such as phenyl oxalate ester (cyalume), to produce chemiluminescence; this application is most commonly encountered in the form of glow sticks.

Horticulture

Some horticulturists and users of hydroponics advocate the use of weak hydrogen peroxide solution in watering solutions. Its spontaneous decomposition releases oxygen that enhances a plant's root development and helps to treat root rot (cellular root death due to lack of oxygen) and a variety of other pests.^{[87][88]}

For general watering concentrations around 0.1% is in use and this can be increased up to one percent for anti-fungal actions.^[89] Tests show that plant foliage can safely tolerate concentrations up to 3%.^[90]



Chemiluminescence of cyalume, as found in a glow stick

Fishkeeping

Hydrogen peroxide is used in aquaculture for controlling mortality caused by various microbes. In 2019, the U.S. FDA approved it for control of *Saprolegniasis* in all coldwater finfish and all fingerling and adult coolwater and warmwater finfish, for control of external columnaris disease in warm-water finfish, and for control of *Gyrodactylus* spp. in freshwater-reared salmonids.^[91] Laboratory tests conducted by fish culturists have demonstrated that common household hydrogen peroxide may be used safely to provide oxygen for small fish. The hydrogen peroxide releases oxygen by decomposition when it is exposed to catalysts such as manganese dioxide.

Removing yellowing from aged plastics

Hydrogen peroxide may be used in combination with a UV-light source to remove yellowing from white or light grey acrylonitrile butadiene styrene (ABS) plastics to partially or fully restore the original color. In the retrocomputing scene, this process is commonly referred to as retrobright.

Safety

Regulations vary, but low concentrations, such as 5%, are widely available and legal to buy for medical use. Most over-the-counter peroxide solutions are not suitable for ingestion. Higher concentrations may be considered hazardous and typically are accompanied by a safety data sheet (SDS). In high concentrations, hydrogen peroxide is an aggressive oxidizer and will corrode many materials, including human skin. In the presence of a reducing agent, high concentrations of H_2O_2 will react violently.^[92] While concentrations up to 35% produce only "white" oxygen bubbles in the skin (and some biting pain) that disappear with the blood within 30–45 minutes, concentrations of 98% dissolve paper. However concentrations as low as 3% can be dangerous for the eye because of oxygen evolution within the eye.^[93]

High-concentration hydrogen peroxide streams, typically above 40%, should be considered hazardous due to concentrated hydrogen peroxide's meeting the definition of a DOT oxidizer according to U.S. regulations, if released into the environment. The EPA Reportable Quantity (RQ) for D001 hazardous wastes is 100 pounds (45 kg), or approximately 10 US gallons (38 L), of concentrated hydrogen peroxide.



Skin shortly after exposure to 35% H_2O_2

Hydrogen peroxide should be stored in a cool, dry, well-ventilated area and away from any flammable or combustible substances. It should be stored in a container composed of non-reactive materials such as stainless steel or glass (other materials including some plastics and aluminium alloys may also be suitable).^[94] Because it breaks down quickly when exposed to light, it should be stored in an opaque container, and pharmaceutical formulations typically come in brown bottles that block light.^[95]

Hydrogen peroxide, either in pure or diluted form, may pose several risks, the main one being that it forms explosive mixtures upon contact with organic compounds.^[96] Distillation of hydrogen peroxide at normal pressures is highly dangerous. It is also corrosive, especially when concentrated, but even domestic-strength solutions may cause irritation to the eyes, mucous membranes, and skin.^[97] Swallowing hydrogen peroxide solutions is particularly dangerous, as decomposition in the stomach releases large quantities of gas (ten times the volume of a 3% solution), leading to internal bloating. Inhaling over 10% can cause severe pulmonary irritation.^[98]

With a significant vapour pressure (1.2 kPa at 50 °C),^[99] hydrogen-peroxide vapour is potentially hazardous. According to U.S. NIOSH, the immediately dangerous to life and health (IDLH) limit is only 75 ppm.^[100] The U.S. Occupational Safety and Health Administration (OSHA) has established a permissible exposure limit of 1.0 ppm calculated as an 8-hour time-weighted average (29 CFR 1910.1000, Table Z-1).^[96] Hydrogen peroxide also has been classified by the American Conference of Governmental Industrial Hygienists (ACGIH) as a "known animal carcinogen, with unknown relevance on humans".^[101] For workplaces where there is a risk of exposure to the hazardous concentrations of the vapours, continuous monitors for hydrogen peroxide should be used. Information on the hazards of hydrogen peroxide is available from OSHA^[96] and from the ATSDR.^[102]

Wound healing

Historically, hydrogen peroxide was used for disinfecting wounds, partly because of its low cost and prompt availability compared to other antiseptics.^[103]

There is conflicting evidence on hydrogen peroxide's effect on wound healing. Some research finds benefit, while other research find delays and healing inhibition.^[104] Its use for home treatment of wounds is generally contraindicated.^[105] 1.5–3% Hydrogen peroxide is used as a desinfectant in dentistry, especially in endodontic treatments together with hypochlorite and chlorhexidin and 1–1.5% is also useful for treatment of inflammation of third molars (wisdom teeth).^[106]

Use in alternative medicine

Practitioners of alternative medicine have advocated the use of hydrogen peroxide for various conditions, including emphysema, influenza, AIDS, and in particular cancer.^[107] There is no evidence of effectiveness and in some cases it has proved fatal.^{[108][109][110][111][112]}

Both the effectiveness and safety of hydrogen peroxide therapy is scientifically questionable. Hydrogen peroxide is produced by the immune system, but in a carefully controlled manner. Cells called phagocytes engulf pathogens and then use hydrogen peroxide to destroy them. The peroxide is toxic to both the cell and the pathogen and so is kept within a special compartment, called a phagosome. Free hydrogen peroxide will damage any tissue it encounters via oxidative stress, a process that also has been proposed as a cause of cancer.^[113] Claims that hydrogen peroxide therapy increases cellular levels of oxygen have not been supported. The quantities administered would be expected to provide very little additional oxygen compared to that available from normal respiration. It is also difficult to raise the level of oxygen around cancer cells within a tumour, as the blood supply tends to be poor, a situation known as tumor hypoxia.

Large oral doses of hydrogen peroxide at a 3% concentration may cause irritation and blistering to the mouth, throat, and abdomen as well as abdominal pain, vomiting, and diarrhea.^[108] Ingestion of hydrogen peroxide at concentrations of 35% or higher has been implicated as the cause of numerous gas embolism events resulting in hospitalisation. In these cases, hyperbaric oxygen therapy was used to treat the embolisms.^[114]

Intravenous injection of hydrogen peroxide has been linked to several deaths.^{[110][111][112]} The American Cancer Society states that "there is no scientific evidence that hydrogen peroxide is a safe, effective, or useful cancer treatment."^[109] Furthermore, the therapy is not approved by the U.S. FDA.

Historical incidents

- On 16 July 1934, in Kummersdorf, Germany, a propellant tank containing an experimental monopropellant mixture consisting of hydrogen peroxide and ethanol exploded during a test, killing three people.^[115]
- During the Second World War, doctors in German concentration camps experimented with the use of hydrogen peroxide injections in the killing of human subjects.^[116]
- In April 1992, an explosion occurred at the hydrogen peroxide plant at Jarrie in France, due to technical failure of the computerised control system and resulting in one fatality and wide destruction of the plant.^[117]
- Several people received minor injuries after a hydrogen peroxide spill on board a flight between the U.S. cities of Orlando and Memphis on 28 October 1998.^[118]
- The Russian submarine *K-141 Kursk* sailed to perform an exercise of firing dummy torpedoes at the *Pyotr Velikiy*, a *Kirov*-class battlecruiser. On 12 August 2000, at 11:28 local time (07:28 UTC), there was an explosion while preparing to fire the torpedoes. The only credible report to date is that this was due to the failure and explosion of one of the Kursk's hydrogen peroxide-fueled torpedoes. It is believed that HTP, a form of highly concentrated hydrogen peroxide used as propellant for the torpedo, seeped through its container, damaged either by rust or in the loading procedure back on land where an incident involving one of the torpedoes accidentally touching ground went unreported. The vessel was lost with all hands. A similar incident was responsible for the loss of *HMS Sidon* in 1955.^[119]
- On 15 August 2010, a spill of about 30 US gallons (110 L) of cleaning fluid occurred on the 54th floor of 1515 Broadway, in Times Square, New York City. The spill, which a spokesperson for the New York City fire department said was of hydrogen peroxide, shut down Broadway between West 42nd and West 48th streets as fire engines responded to the hazmat situation. There were no reported injuries.^[120]

See also

- FOX reagent, used to measure levels of hydrogen peroxide in biological systems

- Hydrogen chalcogenide
- Retrobright, a process using hydrogen peroxide to restore yellowed Acrylonitrile butadiene styrene plastic

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24. See, for example:

- In 1882, Kingzett proposed as a structure $\text{H}_2\text{O}=\text{O}$. See: Kingzett T (29 September 1882). "On the activity of oxygen and the mode of formation of hydrogen dioxide" (<https://books.google.com/books?id=1OYEAAAAQAAJ&pg=RA1-PA141>). *The Chemical News*. **46** (1192): 141–142. Archived (<https://web.archive.org/web/20160903031312/https://books.google.com/books?id=1OYEAAAAQAAJ&pg=RA1-PA141>) from the original on 3 September 2016. Retrieved 9 February 2016.
 - In his 1922 textbook, Joseph Mellor considered three hypothetical molecular structures for hydrogen peroxide, admitting (p. 952): "... the constitution of this compound has not been yet established by unequivocal experiments". See: Joseph William Mellor, *A Comprehensive Treatise on Inorganic and Theoretical Chemistry*, vol. 1 (London, England: Longmans, Green and Co., 1922), p. 952–956. (<https://books.google.com/books?id=PnjVAAAAMAAJ&pg=PA952>) Archived (<https://web.archive.org/web/20160903145901/https://books.google.com/books?id=PnjVAAAAMAAJ&pg=PA952>) 3 September 2016 at the [Wayback Machine](#)
 - W. C. Schumb, C. N. Satterfield, and R. L. Wentworth (1 December 1953) "Report no. 43: Hydrogen peroxide, Part two" (https://hydrogen-peroxide.us/chemical-properties/MIT-Schumb_et_al-hydrogen_Peroxide_ch5_n_ch6-1954.pdf) Archived (https://web.archive.org/web/20150226181901/https://hydrogen-peroxide.us/chemical-properties/MIT-Schumb_et_al-hydrogen_Peroxide_ch5_n_ch6-1954.pdf) 26 February 2015 at the [Wayback Machine](#), Office of Naval Research, Contract No. N5ori-07819 On p. 178, the authors present six hypothetical models (including cis-trans isomers) for hydrogen peroxide's molecular structure. On p. 184, the present structure is considered almost certainly correct—although a small doubt remained. (Note: The report by Schumb et al. was reprinted as: W. C. Schumb, C. N. Satterfield, and R. L. Wentworth, *Hydrogen Peroxide* (New York, New York: Reinhold Publishing Corp. (American Chemical Society Monograph), 1955).)
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External links

- [Hydrogen Peroxide \(https://www.periodicvideos.com/videos/mv_hydrogen_peroxide_accident.htm\)](https://www.periodicvideos.com/videos/mv_hydrogen_peroxide_accident.htm) at *The Periodic Table of Videos* (University of Nottingham)
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